Normal Force Does Not Always Equal Weight

SYNOPSIS

Every scale on Earth measures the *normal force* — how hard the surface pushes back on you — not your true weight. The two agree only when you are not accelerating. In this note you will alter a scale's reading with your own motion, record the normal force with a Vernier force platform while riding an elevator, and explain every feature of that graph with Newton's second law.

BEFORE YOU READ ON — PREDICT

You step on a bathroom scale in an elevator on the second floor and ride down to the ground floor, where the elevator comes to a stop. During that ride down, does the scale ever read *more* than your true weight?

Write down yes or no, and one sentence saying why, before you turn the page. You will check it against the recorded graph.

PERFORMANCE EXPECTATIONS

1. **Apply** ΣF = ma to an object on a scale, with up as positive, and solve for the normal force.
2. **State**, for a given phase of an elevator ride, whether the scale reads more than, less than, or equal to *mg*, and which way the acceleration points.
3. **Read** a normal-force-versus-time graph and label each plateau with the motion that produced it.
4. **Explain** why a scale reads zero in free fall.
5. **Determine** the normal force on an object on a level surface when an applied force is directed at an angle.

The four experiments

1. **Spring scale:** hang a mass, then accelerate the scale up and down. Watch the reading at every phase — starting, moving, stopping.
2. **Digital scale, tilted:** the reading drops as the angle steepens. Your free-body diagram explains why — the scale reads a force component, not a mass.
3. **Bathroom scale squats:** stand still, then squat down and up. Which way does the needle deflect at each phase of the motion, and what does that tell you?
4. **The elevator ride — your data:** you ride with the Vernier force platform under your feet, starting from the second floor and descending. It traces normal force vs. time for the whole trip. Every bump and dip in that graph is Newton's second law speaking, and you will plot and tabulate it yourself below.

The key relationships

Eq. 1 $N = mg$

at rest *or* at constant velocity — the only case where the scale shows true weight

Eq. 2 $N = m(g + a)$

a is signed: positive when it points up (scale reads heavy), negative when it points down (light)

Eq. 3 $N = 0 \;\; \text{when} \;\; a = \, -g$

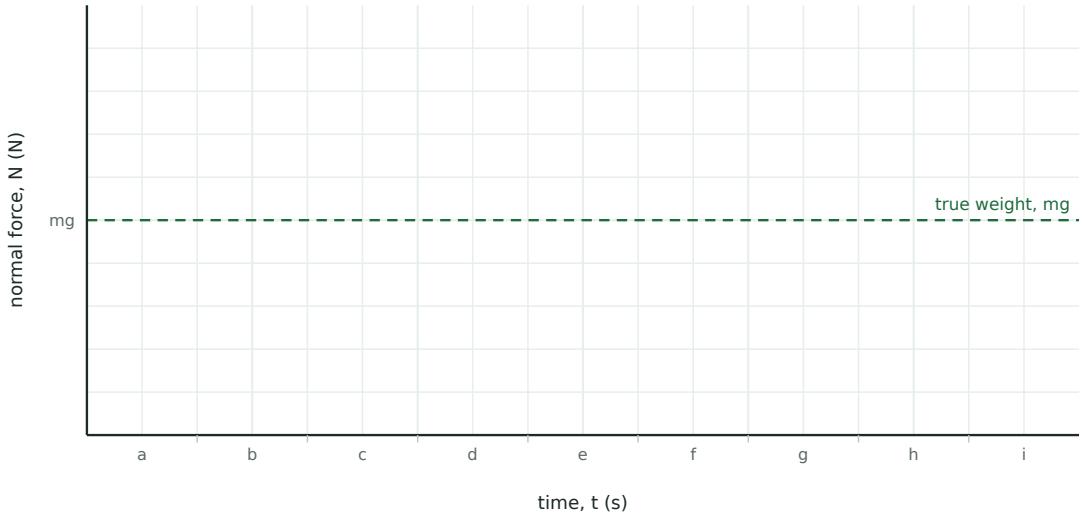
free fall — apparent weightlessness, the scale reads zero

Eq. 4 $N = mg \pm F\sin\theta$

level surface, no vertical acceleration, applied force at angle θ: plus when you push down on it, minus when you pull up

Record your ride

You will ride the elevator with the force platform under your feet, from the second floor down to the ground floor and back. The meter traces the normal force against time for the whole trip. Before you look at the trace, **sketch what you expect** on the axes below; then draw what the meter actually recorded on top of it, and mark where the two differ.



Normal force vs. time for one full trip. The dashed line is your true weight, *mg* — measure it standing still before you start. The letters a–i mark the nine phases of the ride listed in the table below.

Tabulate what you recorded

Work from your own trace. Fill in the last three columns: what the scale read, which way the acceleration pointed during that phase, and its value.

PHASE	WHAT THE ELEVATOR IS DOING	SCALE READS	DIRECTION OF A	A (M/S²)
a	at rest on the second floor			
b	starts moving down — speeding up			
c	moving down at constant speed			
d	still moving down, slowing to a stop			
e	at rest on the ground floor			
f	starts moving up — speeding up			
g	moving up at constant speed			
h	still moving up, slowing to a stop			
i	back at rest on the second floor			

The rule that explains the whole graph

REMEMBER THIS ONE

The sign of a is set by whether you are **speeding up or slowing down** — not by which way you are moving.

Free fall: the scale reads zero

BACK TO YOUR PREDICTION

You predicted whether the scale ever reads more than your true weight during a ride down. Was it right? Name the phase of the ride that settles it, and say which way a points there.

WATCH · THE NOTE WORKED THROUGH

Recording to come. Print the guided notes (link just below), watch, and fill them in as you go.

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